

sTMS Interface Functional Specification

Institute of Technical Education – Course and
Module Setup Information

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/Updated By
0.1	28/05/2022	Initial version of ITE IFS	AvePoint
0.2	22/06/2022	Update based on ITE's feedback on 13 June 2022 and email feedback	AvePoint
0.3	08/07/2022	Update scenario based on discussion with ITE on 5 July 2022	AvePoint
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0.5	25/08/2022	Update based on discussion and email feedback 1. Update csv file names in chapter 2. 2. Remove chapter 3.3 Certificate Fields. 3. Update "Version" field from mandator to non-mandator in chapter 3. 4. Remove "TPGNumber", "TPGApproved", "TPGCouseReferenceld", "ModifiedTime", "ApprovalDate", "TrainingHours" fields in chapter 3.	AvePoint
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Document Acceptance Form

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Institute of Technical Education iStudent system.

ITE iStudent system shall generate CSV files containing the course and module records, and the CSV files shall be encrypted by the PGP protocol and pushed to the sTMS SFTP server accordingly.

1.2 Intended Audience

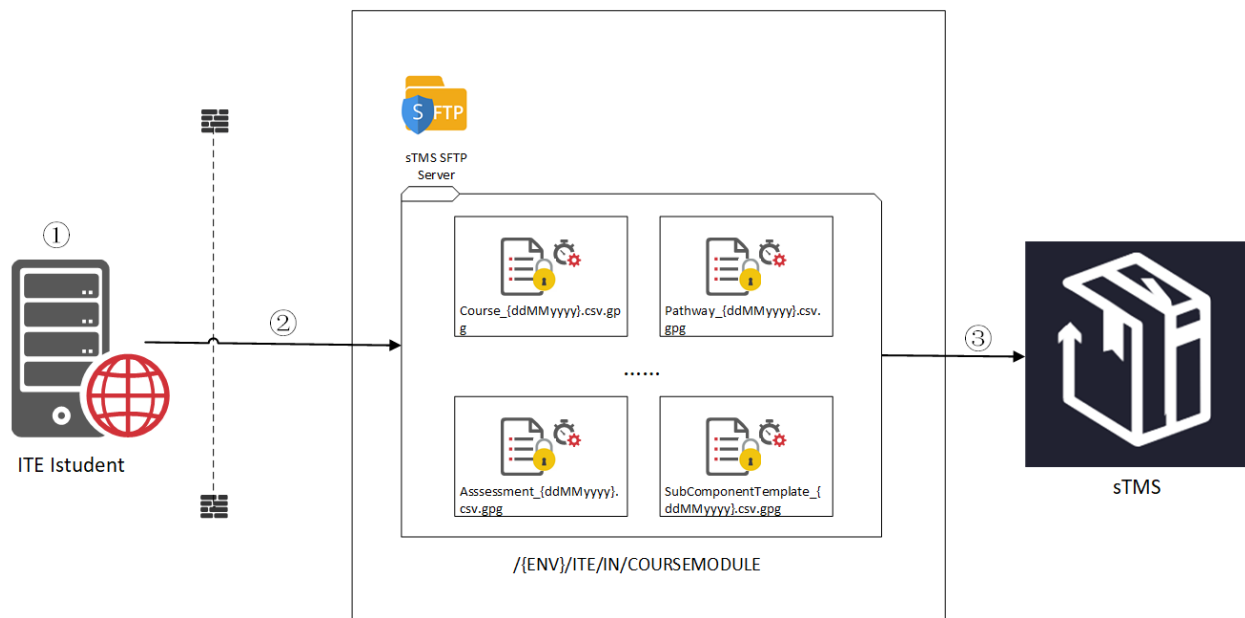
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **ITE iStudent System:** Institute of Technical Education iStudent system.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores CSV files used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	ITE iStudent System generates CSV files based on a pre-defined schedule.
2	ITE iStudent System pushes the CSV files into sTMS SFTP server after encryption with PGP protocol.
3	sTMS pull the CSV files on a pre-defined schedule and retrieve the records after decryption.

2.2 Interface Design Summary

Output Interface File

Interface File Name	Course_{ddMMyyyy}.csv.gpg
---------------------	---------------------------

	Pathway_{ddMMyyyy}.csv.gpg PathwayStructure_{ddMMyyyy}.csv.gpg Module_{ddMMyyyy}.csv.gpg ModuleCertificate_{ddMMyyyy}.csv.gpg Assessment_{ddMMyyyy}.csv.gpg AssessmentComponent_{ddMMyyyy}.csv.gpg AssessmentSubComponent_{ddMMyyyy}.csv.gpg SubComponentTemplate_{ddMMyyyy}.csv.gpg
Data Source System	Institute of Technical Education iStudent System
Data Destination System	shared Training Management System
Mode of Interface	CSV file
Method	SFTP
File Path	{ENV}/ITE/IN/COURSE_MODULE ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	The interface will conduct full-load synchronization.
Frequency	Daily
Timing	01:00am
File Retention Policy	Files are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted.
Possible Exception and Exception Handling	1. File does not exist in the SFTP folder. Exception handling steps: - sTMS and ITE support team shall be informed by email of the file read failure. - iStudent redeposits the files in SFTP. 2. Data verification failed. Exception handling steps: - sTMS support team and ITE user shall be informed by email of the transfer failure. - AP team checks the relevant logs and files to troubleshoot the cause of the failure (data error types are listed below). ❖ Field exceeds max length allowed ❖ Field is mandatory but missing ❖ Key value cannot be found in sTMS (E.g. Grading scheme sync from iStudent not exist in IHL master data) - iStudent needs to fix the issue and redeposits the correct files in SFTP.

2.3 Business Rule

The iStudent system would create and update course information. sTMS would ensure that all new records and changes are identified and stored.

For any changes in course information in iStudent, sTMS would identify the records based on the same course codes and effective dates to update the changes. The Effective Date would be used as the primary key to ensure all of the courses can be matched and synced.

Interface only sync the MOE Academic Courses, other course category and course type data will not be created in sTMS through interface.

iStudent will be the single source of truth for the fields can sync over to sTMS, otherwise the Default to Null or Set value can still updated by user in sTMS.

Below section will describe the course version creation, update and deactivation business rules. For course synchronization scenario details, please refer to the following document.



sTMS-ITE-Course
Interfacing.xlsx

2.3.1 Create Course Version Business Rule

The create course version business rules are described below:

Scenario 1: When a course record is created on the iStudent system, and this record "Effective Date" later than other records, sTMS will create a new course version for this course and update the previous version "Effective Date To" value to the day before the new version "Effective Date From" date. There shall be no gap between the course versions date.

Scenario 1

iStudent

create new version at the end

input file syncing date: 2022-05-01

before

Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Active

after

Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Active
C01	2022-10-01	Active

STMS

before

Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01		3	Active

after

Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01		4	Active

Syncing

Scenario 2: When a course record is created to the iStudent system, and the "Effective Date" of the created course is between the initial effective dates, sTMS shall create a new version for this record, set "Effective Date To" one day before next effective version "Effective Date From", and update the previous effective version "Effective Date To" one day before the newly added version.

Scenario 2				sTMS				
create new version in between								
input file syncing date: 2022-05-01								
Before				Before				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active
C01	2022-10-01	Active		C01	2022-10-01		3	Active
After				After				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01	2022-07-31	2	Active
C01	2022-08-01	Active		C01	2022-10-01		3	Active
C01	2022-10-01	Active		C01	2022-08-01	2022-09-30	4	Active

Scenario 3: A scenario is considered where the user would like to put a stop to the course, an "Inactive" course record would be created in iStudent, the "Effective Date To" for the previous effective version would be updated.

Scenario 3				sTMS				
create an Inactive version								
input file syncing date: 2022-05-01								
Before				Before				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01		2	Active
After				After				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active
C01	2022-10-01	Inactive						

Scenario 4: When a previous record created in iStudent system is marked as Inactive and the user created a new course version is created later than the "Inactive" record, sTMS would create a new course version with "Active" status and leaving the "Effective Date To" as empty.

Scenario 4				sTMS				
create new version after Inactive version								
input file syncing date: 2022-05-01								
Before				Before				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active
C01	2022-10-01	Inactive						
After				After				
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active
C01	2022-10-01	Inactive						
C01	2022-11-01	Active		C01	2022-11-01		3	Active

Scenario 5: When the previous course record is "Inactive" and user creates a new "Active" record in between in iStudent, sTMS would capture the "Active" record as a new course version.

Scenario 5			iStudent			sTMS		
			create an active version after an inactive version in between					
			input file syncing date: 2022-05-01					
			Before			Before		
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active	
C01	2022-06-01	Active	C01	2022-06-01	2022-09-30	2	Active	
C01	2022-10-01	Inactive	C01	2022-12-01		3	Active	
C01	2022-12-01	Active						
			After			After		
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active	
C01	2022-06-01	Active	C01	2022-06-01	2022-09-30	2	Active	
C01	2022-10-01	Inactive	C01	2022-12-01		3	Active	
C01	2022-11-01	Active	C01	2022-11-01	2022-11-30	4	Active	
C01	2022-12-01	Active						

2.3.2 Update Course Version Business Rule

If the "Effective Date" of the updated course record did not change in iStudent, sTMS will update the data of the corresponding version, which will have no impact to existing course version periods.

When a record needs to be updated in sTMS, if The Allow Update column of the field is Y, the value of the original field will be replaced with the value of the field of the new record. If The Allow Update column of the field is N, the value of the original field will not be changed. Please refer to [3. Interface fields](#) for more field-level details.

The update course version business rules are described below:

Scenario 1&2: If the "Effective Date" changes in iStudent, sTMS will follow the delete course version business rule to deactivate the corresponding course version and follow the create course version business rule to create a new course version for the updated course records.

Scenario 1			iStudent			sTMS		
			Update the last version					
			input file syncing date: 2022-05-01					
			Before			Before		
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active	N
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active	N
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active	N
C01	2022-10-01	Active	C01	2022-10-01		4	Active	N
			After			After		
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active	N
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active	N
C01	2022-07-01	Active	C01	2022-07-01	2022-11-30	3	Active	N
C01	2022-12-01	Active	C01	Set to an earlier date		4	Active	Y
			C01	2022-12-01		5	Active	N

Scenario 3	iStudent	sTMS
Update the last version from inactive to active		
input file syncing date: 2022-05-01		
Before		Before
Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Active
C01	2022-10-01	Inactive
After		After
Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Active
C01	2022-10-01	Active

Scenario 4 iStudent

Update version from inactive to active in between

input file syncing date: 2022-05-01

sTMS

Before		
Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Inactive
C01	2022-10-01	Active



Syncing

Before				
Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	2022-09-30	3	Inactive
C01	2022-10-01		4	Active

After		
Course Code	Effective Date	iStudent Status
C01	2022-01-01	Active
C01	2022-04-01	Active
C01	2022-07-01	Active
C01	2022-10-01	Active

After				
Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	2022-09-30	3	Inactive
C01	2022-10-01		4	Active
C01	2022-07-01	2022-09-30	5	Active

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Scenario 5 **iStudent** **sTMS**

update an Inactive version effective date
input file syncing date: 2022-05-01

Before			Before				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active	C01	2022-06-01	2022-09-30	2	Active
C01	2022-10-01	Inactive					

Syncing

After			After				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active
C01	2022-06-01	Active	C01	2022-06-01	2022-10-31	2	Active
C01	2022-11-01	Inactive					

2.3.3 Deactivate Course Version Business Rule

The deactivated course version business rules are described below:

Scenario 1: When a course record is deactivated in the iStudent system, and this record is the last version in sTMS, sTMS will mark the corresponding deactivate course version as inactive and retain the "Effective Date To" of the previous version.

Scenario 1 **iStudent** **sTMS**

deactivate the last version
input file syncing date: 2022-05-01

before			before				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01	Active	C01	2022-10-01		4	Active

Syncing

After			After				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01	Inactive	C01	2022-10-01		4	Inactive

Scenario 2: If a course record is deactivated in the iStudent system, and the "Effective Date" of the created course is between the initial effective dates, sTMS will mark the corresponding course version as inactive and retain the previous version "Effective Date To" date.

Scenario 2 **iStudent** **sTMS**

deactivate version in between
input file syncing date: 2022-05-01

Before			Before				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01	Active	C01	2022-10-01		4	Active

Syncing

After			After				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS status
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Inactive	C01	2022-07-01	2022-09-30	3	Inactive
C01	2022-10-01	Active	C01	2022-10-01		4	Active

Scenario 3: If a course is deactivated in the iStudent system, and the inactive course is the past version in sTMS, sTMS shall update the course version as Inactive.

Scenario 3
deactivate past version in between
input file syncing date: 2022-05-01

before			before				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS status
C01	2022-01-01	Active	C01	2022-01-01	2022-03-31	1	Active
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01	Active	C01	2022-10-01		4	Active

Syncing

After			After				
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS status
C01	2022-01-01	Inactive	C01	2022-01-01	2022-03-31	1	Inactive
C01	2022-04-01	Active	C01	2022-04-01	2022-06-30	2	Active
C01	2022-07-01	Active	C01	2022-07-01	2022-09-30	3	Active
C01	2022-10-01	Active	C01	2022-10-01		4	Active

2.3.4 Delete Course Version Business Rule

If iStudent deletes an inactive record, sTMS will set a backend delete indicator to "Y" to ensure sure no time stamp gap between active course versions.

The delete course version business rules are described below:

Scenario 1: When a course record is deleted in the iStudent system, and the "Effective Date" of the deleted course is between the initial effective dates, sTMS will mark the corresponding course delete indicator as "Y" and update the previous effective version "Effective Date To" one day before the newly version.

Scenario 1
delete Inactive version in between
input file syncing date: 2022-05-01

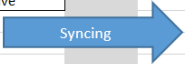
Before			Before					
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active	N
C01	2022-06-01	Active	C01	2022-06-01	2022-09-30	2	Active	N
C01	2022-10-01	Inactive	C01	2022-10-01	2022-10-31	3	Inactive	N
C01	2022-11-01	Active	C01	2022-11-01		4	Active	N

Syncing

After			After					
Course Code	Effective Date	iStudent Status	Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active	C01	2022-01-01	2022-05-31	1	Active	N
C01	2022-06-01	Active	C01	2022-06-01	2022-10-31	2	Active	N
C01	2022-10-01	Inactive	C01	2022-10-01	2022-10-31	3	Inactive	Y
C01	2022-11-01	Active	C01	2022-11-01		4	Active	N

Scenario 2: When a course record is deleted in the iStudent system, and this record is the last version in sTMS, sTMS will mark the corresponding course delete indicator as "Y" and clean the previous effective version "Effective Date To".

Scenario 2				sTMS					
iStudent									
delete the last version									
input file syncing date: 2022-05-01									
Before				Before					
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active	N
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active	N
C01	2022-10-01	Active		C01	2022-10-01	2022-10-31	3	Active	N
C01	2022-11-01	Inactive		C01	2022-11-01		4	Inactive	N
After				After					
Course Code	Effective Date	iStudent Status		Course Code	Effective Date From	Effective Date To	Version Number	sTMS Status	Delete Indicator
C01	2022-01-01	Active		C01	2022-01-01	2022-05-31	1	Active	N
C01	2022-06-01	Active		C01	2022-06-01	2022-09-30	2	Active	N
C01	2022-10-01	Active		C01	2022-10-01		3	Active	N
C01	2022-11-01	Inactive		C01	2022-11-01		4	Inactive	Y



3. Interface Fields

3.1 Course Structure Fields

sTMS Field Name	iStudent Entity Name	iStudent Field Name	Description	Data Type	Allow Update	Remarks
Course						
Course_id*	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: module_id (6 character number) FQ/Program course: ACAD_PLAN	The id of the Course	String	N	Sample: BRPRG or 000012
Version*	-	-	The version of course	Integer	-	sTMS will create course version according to business rule
Institution*	ACAD_PLAN_TBL	INSTITUTION	The Institution of course	String	N	Hardcoded as "(ITE) Institute of Technical Education"
Branch	CRSE_OFFER	CAMPUS	The branch of a course	String	N	Sample: HQ#;CC#;CE#;CW
School*	A_CRSE_CATALOG	A_ISCH_TYPE	The school of a course	String	N	Sample: SBS-CC#;SBS-CW#;SBS-CE
TPG Applicable*	-	-	Defines whether the course is approved by TPG	String	-	Default to Null, if TPG course reference No. have value, set to Yes
TPG course reference No.	A_SFC_CRSE_MAP	A_SFC_CRSE_ID	Displayed and editable only when "TPG Approved" is "Yes"	String	N	Sample: SCN-198202248E-01-CRS-N-0027685
Course name*	Normal course: CRSE_CATALOG	Normal course: COURSE_LONG_TITLE	The name of the course. If have TPG No., sync from TPG first.	String	N	Sample: Bridging Programme for Progression to Polytechnic Diploma

	FQ/Program course: ACAD_PLAN_TBL	FQ/Program course: DIPLOMA_DESCR				
Course short name	-	-	The short name of the course	String	-	Default to Null
Course alternative name	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: COURSE_LONG_TITLE FQ/Program course: DIPLOMA_DESCR	The alternative name of the course.	String	Y	Sample: BRPRG
Course code*	Normal course: CRSE_OFFER FQ/Program course: ACAD_PLAN_TBL	Normal course: Course type + SUBJECT + CATALOG_NBR FQ/Program course: ACAD_PLAN	1. The internal course code defined by each IHL 2. Course code should be unique for a course and will be fixed for all versions of a course	String	N	This field is the primary key to make sure all the courses can be matched and synced
Diploma course code	-	-	The ID of diploma for the course	String	-	Default to Null
Effective From*	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: EFFDT FQ/Program course: EFFDT	The effective date from of course	String(\$Date)	Y	This field is the primary key to make sure all the courses can be matched and synced Sample: 28/10/2022
Effective To	-	-	The effective date to of course	String(\$Date)	-	Default to Null. This field sTMS should generate according to business rule
No. of training hours*	Normal course: A_MOD_OFFER_HRS	Normal course: CRSE_CONTACT_HRS	The total training number of hours of a course. If have TPG No., sync from TPG first.	Integer	Y	Sample: 20

	FQ/Program course: NA (Calculate based on module training hour)	FQ/Program course: NA (Calculate based on module training hour)				
No. of assessment hours	-	-	The total assessment number of hours of a course	Integer	-	Default to Null.
Course category*	-	Acad Career	The Course Category for the course	String	N	Refer to 5. Appendix for the mapping. Sample: CET
Course type*	-	Acad Group	The Course Type for the course	String	N	Refer to 5. Appendix for the mapping. Sample: HNCE
Senate reference No.	-	-	The senate reference number for the course.	String	-	-
Approval date*	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: EFFDT FQ/Program course: EFFDT	Mandatory only when "TPG Approved" is "Yes"	String	N	Sample: 28/10/2022
Sunset date	-	-	The date of term end	String(\$Date)	-	Default to Null.
WSP indicator*	-	-	WSP Indicator	String	-	Default to Null. sTMS set value to N/A
National Manpower Cluster	-	-	The National Manpower Cluster for the course	String	-	Default to Null.
Area of Interest*	-	-	The area of interest for the course	String	-	Default to Null, sTMS set value to "Others – Others"

TPG contact	-	-	The information was retrieved from TPG, only requires when "TPG Approved" is "Yes"	String	-	Default to Null. If have TPG No., sync from TPG.
With ceremony*	-	-	1. Define if the course has ceremony, if yes is checked. This course will be loaded in the ceremony module. 2. If no is checked, the course will not be loaded in the ceremony module.	String	-	Default to Null, sTMS set value to N
Add to graduation list*	-	-	1. If Yes is checked, student's enrolment status will become "graduated" after completion of course 2. If No is checked, student's enrolment status will become "completed" after completion of the course.	String	-	Default to Null, Course type is Higher Nitec or Nitec and is Program/FQ course (multi module), or Course type is ISC, sTMS set value to Y, otherwise set to N
TPG contact's email address	-	-	The information was retrieved from TPG, only requires when "TPG Approved" is "Yes"	String	-	Default to Null. If have TPG No., sync from TPG.
TPG contact's phone number	-	-	The information was retrieved from TPG, only requires when "TPG Approved" is "Yes"	String	-	Default to Null. If have TPG No., sync from TPG.
Allow Ranking*	-	-	Allow applicant to rank courses during application so that evaluator can take reference while giving out offers.	String	-	Default to Null, sTMS set value to N
Allow displaying MER check*	-	-	If select Yes, system will perform MER check against application form & their past student record if applicable	String	-	Default to Null, sTMS set value to N

Require emergency contact	-	-	If select Yes, system will require emergency contact	String	-	Default to Null, sTMS set value to N
Course objective	-	-	The information was retrieved from TPG, only requires when "TPG Approved" is "Yes"	String	-	Default to Null. If have TPG No., sync from TPG.
Course description	-	-	The information was retrieved from TPG, only requires when "TPG Approved" is "Yes"	String	-	Default to Null. If have TPG No., sync from TPG.
Status*	-	-	Course setup status	String	-	Default to Null, sTMS set value to Pending fee setup
SSG Related fields (Total 12 fields) all default to Null, E.g., Target audience, Target training group, etc. If have TPG No., sync from TPG.						
Required student accounts	-	-	Required student accounts	String	-	Default to Null
Course owners	-	-	Course owners	String	-	Default to Null
Select a MER to preview	-	-	Select a MER form From MER bank	String	-	Default to Null
Issue course level e-Transcript/SOR	-	-	Specify whether to issue course level e-Transcript/SOR.	String	-	Base on course type, one course one module course type is Nitec, Higher Nitec, sTMS set value to Y, FQ (one course with multi value) set to N, otherwise set to N
Course Level Document Type	-	-	Available when "Issue course level e-Transcript/SOR" is checked.	String	-	When "Issue course level e-Transcript/SOR" is checked. sTMS set value to SOR
Course Level Template	-	-	Available when "Issue course level e-Transcript/SOR" is checked.	String	-	When "Issue course level e-Transcript/SOR" is checked.

						sTMS set value to Nitec and Higher Nitec SOR
Issue Statement of Result (SOR) to student by semester (if any)	-	-	Specify whether to issue semester level e-Transcript/SOR.	String	-	Default to Null
Semester Level Result Type	-	-	Available when "Issue Statement of Result (SOR) to student by semester (if any)" is checked.	String	-	Default to Null
Semester Level Template	-	-	Available when "Issue Statement of Result (SOR) to student by semester (if any)" is checked.	String	-	Default to Null
Worktag Related fields (Total 13 fields) only sync AMC, otherwise default to Null						
AMC	A_CRSE_CATALOG	COMMITTEE_ID	-	String	Y	
Pathway						
Course_id*	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: module_id (6 character number) FQ/Program course: ACAD_PLAN	The id of the Course	String	N	Sample: BRPRG or 000012
Pathway_id*	-	Normal course: module_id + "_1"(6 character number) FQ/Program course:	The id of the Pathway	String	N	Sample: BRPRG_1 or 000012_1

		ACAD_PLAN + "_1"				
Pathway Type*	-	-	The pathway type of the course	String	-	Default to Null, sTMS set to Normal Pathway
Pathway Name*	RQ_GRP_TBL	DESCR254A	The pathway name of the course	String	Y	Sample: BRPRG_Bridging Prog for Poly Diploma Requirements.
Minimum Requirement Type*	-	-	The minimum entry requirement of the course, the value should be "By course number" if the pathway type is "Programme Pathway"	String	-	Default to Null, sTMS set to By modular credit number
Number of elective*	-	-	The minimum number of elective module/credit/course for the pathway	Integer	-	Default to Null, sTMS set to 0
PathwayStructure						
Pathway_id*	-	Normal course: module_id + "_1"(6 character number) FQ/Program course: ACAD_PLAN + "_1"	The id of the Pathway	String	N	Sample: BRPRG_1 or 000012_1
Module/Course ID*	CLST_DETL_TBL	CRSE_ID	The module code of the module	String	N	Sample: 000012
Type*	-	-	The type of the Pathway Structure	String	-	Default to Null, sTMS set to Core
CourseCertification (No need to provide this sheet, value will set by sTMS)						
Course_id*	Normal course: CRSE_CATALOG FQ/Program course: ACAD_PLAN_TBL	Normal course: module_id (6 character number)	The id of the Course	String	N	Sample: BRPRG or 000012

		FQ/Program course: ACAD_PLAN				
Certification name*	-	-	The Certificate name of the course	String	N	FQ/Program course (one course multi module) and course type is Higher Nitec or Nitec, set value refer to 5. Appendix For ISC set value to Full Certificate ISC
Certification level*	-	-	The Certificate level of the course	String	N	Default to blank and sTMS set to Course Certificate
Pathway to complete	-	-	The IDs of pathway the certificate is assigned to	String	N	Default to blank and sTMS set to Select All
Module to complete	-	-	The IDs of module the certificate is assigned to	String	N	Default to Null
Number of elective certificate	-	-	Available only when "Course Certificate" is selected as Certificate level	String	N	Default to Null
Certificate type*	-	-	The Certificate type of the course	String	N	Default to Null and sTMS set to Core
WSP Indicator	-	-	WSP Indicator. Only available for normal pathway and Available only when "Post Diploma Certificate" or "Modular Certificate" is selected as Certificate level	String	N	Default to Null and sTMS set to N/A
TPG course reference No.	-	-	TPG course reference No.	String	N	Default to Null

The asterisk * represents the field is required.

3.2 Module Structure Fields

sTMS Field Name	iStudent Entity Name	iStudent Field Name	Description	Data Type	Allow Update	Remarks
Module						
Module_id*	CRSE_CATALOG	CRSE_ID	The id of the module (6 character number)	String	N	Sample: 000006
Version	-	-	The version of module	Integer	-	Default to Null. Generate by sTMS
Institution	-	-	The Institution of course	String	-	Default to (ITE) Institute of Technical Education
Branch	CRSE_OFFER	CAMPUS	The branch of a module	String	N	Sample: HQ#;CC#;CE#;CW
School*	A_CRSE_CATALOG	A_ISCH_TYPE	The school of a module	String	N	Sample: SBS-CC#;SBS-CW#;SBS-CE
Module code*	CRSE_OFFER	SUBJECT+CATALOG_NBR	1. The internal module code defined by each IHL. 2. Module code should be unique for a module and will be fixed for all versions of a module.	String	N	This field is the primary key to make sure all the modules can be matched and synced Sample: AC49020
Module name*	CRSE_CATALOG	COURSE_LONG_TITLE	The name of a module	String	Y	Sample: Business Statistics I
Effective from*	CRSE_CATALOG	EFFDT	The effective date from of module offering Format: DDMMYYYY	String(\$Date)	Y	This field is the primary key to make sure all the modules can be matched and synced Sample: 20102022
Effective to	-	-	The effective date to of module offering	String(\$Date)	-	Default to Null. This field sTMS should generate according to business rule
No. of module hours*	A_MOD_OFFER_HRS	CRSE_CONTACT_HRS	The total number of hours of a module	Integer	N	Sample: 20
No. of modular credits	CRSE_CATALOG	UNITS_MAXIMUM	The number of modular credits of a module	Integer	N	Sample: 4

Module synopsis	-	-	-	String	-	Default to Null
Remarks	-	-	-	String	-	Default to Null
Status*	CRSE_CATALOG	Status	Status of the module	String	Y	Default to Active if no data sync over to sTMS Value list: Active, Inactive
Moudule owners	-	-	-	String	-	Default to Null
Co-requisites	-	-	-	String	-	Default to Null
Equivalencies	CRSE_CATALOG	EQUIV_CRSE_ID	The equivalencies of a module	String	Y	Sample: 000006#;000007
Issue along with course certificate	-	-	-	String	-	Default to Null, sTMS set value to N
Follow service type settings	-	-	-	String	-	Default to Null, sTMS set value to Y
Minimum attendance requirement	-	-	-	Integer	-	Default to Null
Notification threshold	-	-	-	Integer	-	Default to Null
Online Test/Exam	A_MOD_ONLIN_TST	A_ONLINE_IND	-	String	Y	Sample: Y
Module Certification						
master_module_id*	DESCR1	CRSE_ID	The id of the module	String	N	Sample: 000006
Tier*	-	-	-	String	-	Default to Null, sTMS set value to Tier 1
Certificate*	-	Acad Group	The Certificate name of the module	String	Y	Base on course type, sTMS set value refer to 5. Appendix
Attendance	-	-	The module certificate requires min. attendance requirement	String	-	Default to Null, sTMS set value refer to 5. Appendix

Assessment	-	-	The module certificate requires pass of the assessment.	String	-	Default to Null, sTMS set value refer to 5. Appendix
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The asterisk * represents the field is required.

3.3 Assessment Structure Fields

sTMS Field Name	iStudent Entity Name	iStudent Field Name	Description	Data Type	Allow Update	Remarks
Assessment						
Module id*	LAM_CRS_H_SCL	Module ID	The ID of the module	String	N	Sample: 201947
Grade mapping rule*	LAM_CRS_H_SCL	Grading	The name of the grade mapping rule	String	Y	Sample: GSP
Status	-	-	-	String	-	Default to Null, sTMS set value to Completed
Assessment status	-	-	-	String	-	Default to Null, sTMS set value to a list provided by ITE
Allow grade review	-	-	-	String	-	Default to Null, sTMS set value to Y
AssessmentComponent						
Component name*	A_CRSE_COMP_LAM	Category	The name of the component	String	N	Sample: IMPR
Component type	LAM_TYPE	Descr	The type of the component	String	Y	Sample: In-Module (Practical)
Component weightage	-	-	-	-	-	Default to Null, sTMS will auto calculate it by using Sub-component weightage
Component grade mapping rule	LAM_CRS_H_SCL	Grading	The name of the grade mapping rule	String	Y	Sample: GSP
Compulsory to pass indicator	-	-	-	String	-	Default to Null, sTMS set value to N
Compulsory to pass options	-	-	-	String	-	Default to Null

Attain minimum marks/grade	-	-	-	String	-	Default to Null
AssessmentSetting*	A_CRSE_COMP_LAM	Module ID	The ID of the module	String	N	Sample: 201947
AssessmentSubComponent						
Component name*	A_CRSE_COMP_LAM	Category	The name of the component	String	N	Sample: IMPR
Sub-component name*	A_CRSE_COMP_LAM	Assign Item cd	The name of the sub-component to be added for the component	String	N	Sample: PHT1
Sub-component template name*	-	-	-	String	-	Generate by sTMS
Sub-component type	A_CRSE_COMP_LAM	Descr	The type of the sub-component	String	Y	Sample: Phase Test 1
Base score*	-	-	The base score of the sub-component	String	-	Default to Null, sTMS set value to 100
Include in overall module score*	A_CRSE_COMP_LAM	Required	The sub-component score will be included into the overall module score.	String	Y	Value list: Y, N
Sub-component weightage*	A_CRSE_COMP_LAM	Points	The weightage (%) of the sub-component to be added for the component.	Number	Y	Sample: 14
Compulsory to pass indicator	A_CRSE_COMP_LAM	Complsry Pass	The certain criteria need to be met for the sub-component to pass the whole module.	String	Y	If sub-component first 4 character start with "TEXM" and "PEXM", Set to "Y", otherwise take value from iStudent. Value list: Y, N
Pass critical skill	A_CRSE_COMP_LAM	Pass Critical	Have value when Compulsory to pass indicator is "Y". The checkbox in sTMS will be ticked if "Null" value in ITE.	String	Y	Take value from iStudent when Compulsory to pass indicator is "Y", othwise default to Null
Fulfill task list	A_CRSE_COMP_LAM	Fulfill Task	Have value when Compulsory to pass indicator is "Y".	String	Y	Take value from iStudent when Compulsory to pass

			The checkbox in sTMS will be ticked if "Null" value in ITE.			indicator is "Y", otherwise default to Null
Not having assessment status	-	-	Have value when Compulsory to pass indicator is "Y". The checkbox in sTMS will not be ticked	String	-	Set to Y if "Compulsory to pass" is "Y", otherwise default to Null
Assessment status	-	-	Have value when Compulsory to pass indicator is "Y" and Not having assessment status is "N".	String	-	Set to ABS if "Not having assessment status" is "Y", otherwise default to Null
Attain minimum marks	A_CRSE_COMP_LAM	Minimum Mark	Have value when Compulsory to pass indicator is "Y". Upon select 'attain minimum marks', 'attain minimum grade' will become non-selectable, and vice versa.	Number	Y	Sample: 60
Attain minimum grade	-	-	Have value when Compulsory to pass indicator is "Y". Upon select 'attain minimum marks', 'attain minimum grade' will become non-selectable, and vice versa.	String	-	Default to Null
Non-Redistributable Component	-	-	The sub-component is a non-redistributable component or not.	String	-	Default to Null
Category*	LAM_TYPE	End of Mod Flag	The category type of the sub-component iStudent send value N, sTMS set the value to In-class assessment; iStudent send value Y, sTMS set the value to Standalone exam;	String	Y	Value list: N- In-class assessment, Y - Standalone exam
Assignment deadline	-	-	-	Date	-	Default to Null
Session No	-	-	-	String	-	Default to Null
SubComponentTemplate						

Sub-component name*	A_CRSE_COMP_LAM	Assign Item cd	The name of the sub-component to be added for the component	String	N	Sample: PHT1
Sub-component template name*	-	-	The name of the Sub-component template	String	-	Name convention: If sub-component first 4 character start with "TEXM" and "PEXM", Without allow retest: {Grading} – Not Allow Makeup Otherwise, allow retest: {Grading} - Allow Retest – {Capped Mark} Without allow retest: {Grading} – Not Allow Retest Sample: GS1 – Allow Retest – 60 GS2 - Not Allow Retest
Grade mapping rule*	LAM_CRS_GRD_SCL	Grading	The name of the grade mapping rule	String	Y	Sample: GSP
Assessment status	-	-	The assessment status for the sub-component assessment setup template	String	-	Default to Null, sTMS set a list provided by ITE
Re-assessment	A_ASGN_ITEM_TBL	Allow Retest	The sub-component is eligible for re-assessment or not	String	Y	Value list: Y, N
No. of re-assessment allowed	-	-	The number of re-assessment allowed for the sub-components	Number	-	Default to Null, sTMS set value to 1
Re-assessment cap	A_CRSE_COMP_LAM	Capped Mark	The maximum score (%) student can obtain for the re-assessment.	Number	Y	Sample: 60

Re-assessment cut-off	-	-	The lowest mark student needs to obtain to be eligible to attend the re-assessment.	Number	-	Default to Null
Re-assessment fee	-	-	The fee is applicable to the student for the re-assessment	String	-	Default to Null
Make-up assessment	-	-	The make-up assessment is eligible for the sub-component	String	-	If sub-component first 4 character start with "TEXM" and "PEXM", set value to Null, otherwise set value to Y
No. of make-up assessment allowed	-	-	The number of make-up assessment allowed for the sub-components	Number	-	Default to Null, sTMS set value to 1
Make-up assessment cap	-	-	The maximum score (%) student can obtain for the make-up assessment.	Number	-	Default to Null, sTMS set value to 100
Admin fee name	-	-	The Admin fee to request from the student.	String	-	Default to Null
Charge code	-	-	The charge code of the admin fee.	String	-	Default to Null
Rate charged by	-	-	The rate charged by of the admin fee.	String	-	Default to Null
Amount	-	-	The amount of the admin fee.	Number	-	Default to Null

The asterisk * represents the field is required.

4. Sample Files



Sample file for ITE
Course module setup.

5. Appendix

MOE Indicator	sTMS Course Category	sTMS Course Type	Course Cert	PDC/MC Cert	Module Cert	Attendance	Assessment	IS Student Acad Career	IS Student Acad Group	IS Student Course Category (ACAD_GROUP)
MOE										NCE
MOE Indicator										HN
MOE	MOE Academic Courses	Part-Time Nitec	Full Certificate Nitec, COM	NA	Module Certificate Nitec	N	Y	CET	HNCE	ISC
		Part-Time Higher Nitec	Full Certificate Higher Nitec, COM	NA	Module Certificate Higher Nitec	N	Y	CET	NCE	
		ITE Skills Certificate (ISC)	Full Certificate ISC	NA	NA	N	Y	CET	ISC	